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Recession *Probability* Engine

A machine-learning approach to U.S. business-cycle forecasting

AUTHOR
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PERIOD
1950 – present

VALIDATION
11-fold walk-forward, OOS

STACK
XGBoost · SHAP · FRED API

0.827 *avg.*
AUC, MATURE OOS FOLDS (5–11)

14.2 *months*
AVERAGE ADVANCE WARNING

< 0.1 */yr*
FALSE ALARMS (0 FROM FOLD 7)

5 / 6
OOS RECESSIONS CAUGHT (MATURE FOLDS 5–11)

ABSTRACT

We present a machine-learning system that produces a monthly probability estimate of a U.S. recession occurring within the next 12 months. The architecture addresses three structural challenges of macro forecasting: a sparse positive-class history (~11 NBER-defined recessions since 1950), retroactive label definition, and the severe credibility cost of false alarms. It combines XGBoost trained on 58 engineered macroeconomic features from FRED, strict temporal walk-forward validation, and a sustained-signal output filter with hysteresis. Across the strictly out-of-sample mature folds (1975 to present), the system achieves an average AUC of 0.827, an average advance warning of 14.2 months, and fewer than 0.1 false alarms per year — zero from fold 7 onward. It outperforms the univariate 10Y–2Y yield-curve heuristic on both precision and false-positive suppression.

01
INTRODUCTION

The hardest easy problem in applied ML.

Recession forecasting occupies an unusual position in applied machine learning. The prediction target is rare — roughly 11% of post-war months fall within an NBER-defined recession window — its definition is retroactive, and the asymmetric cost of errors is severe. A false alarm is not merely an incorrect prediction; it is a credibility-destroying event that may cause the next genuine signal to be ignored.

Three structural challenges compound the difficulty:

Label scarcity.
The United States has experienced approximately 11 NBER-defined recessions since 1950 — fewer than 130 positive-class samples across 75 years of monthly data. A dataset size that would be considered trivially small in most ML contexts.

Retroactive definition.
The NBER typically announces recession start dates six to twelve months after the fact. Recent training labels are therefore provisional and subject to revision; models trained on the most current data are learning from unresolved ground truth.

Forward contamination.
A 12-month forward label means a sample from early 2019 carries a recession label of 1 simply because COVID arrived in Q1 2020 — not because of any cyclically learnable pattern in 2019. Without an explicit architectural response, the model learns spurious associations.

> *The objective was not to build the most complex forecasting system, but a credible one — one that catches every major recession with meaningful advance warning, while maintaining a false-alarm rate low enough to preserve operational utility over time.*
>
> — PROJECT THESIS

02
DATA & FEATURES

From sixty raw FRED series to fifty-eight engineered features.

All data are sourced from FRED, the St. Louis Federal Reserve's public macroeconomic database. Approximately 60 raw series are fetched via API, cached locally, and aligned to month-end frequency before feature engineering begins.

TRANSFORMATION PHILOSOPHY

Raw series are never used directly as model inputs. Every feature is derived as one or more of: absolute level, year-over-year percentage change, 3-month percentage change, 12-month absolute change, or a binary threshold flag. This layer normalizes units across heterogeneous series and extracts the rate-of-change signals that lead the underlying levels.

Where a series begins after 1950 — VIX before 1990, JOLTS before 2000 — the corresponding features are left as *proper missing values*. No imputation is applied. The production model handles structural missingness natively, learning to weight features relative to their actual historical availability.

FEATURE GROUPS (DATASET E, PLUS CONFIGURATION)

TABLE 1 — PRODUCTION FEATURE INVENTORY · 58 FEATURES ACROSS 8 ACTIVE DOMAINS

GROUP	FEATURES	AVAILABLE FROM
Core macro & labor	9	1939
Inflation & prices	5	1913
Housing & consumption	11	1950
Yield curve & central bank	5	1953
Credit & corporate stress	10	1919
Market volatility & leading indicators	14	1967
Bank lending standards (SLOOS)	1	1990
CapEx intent	3	1992
Total	**58**	**— 1950 floor**

Eight additional feature groups (~27 features) are available under **--feature-set full** *but were excluded from production after seeing a drop in performance when used*

CLASS IMBALANCE

With roughly 11% recession-month prevalence, the dataset is severely imbalanced. Synthetic oversampling (SMOTE) is explicitly rejected: generating artificial recession months would corrupt the temporal structure of the data and produce training observations that violate chronological ordering. Instead, XGBoost's native gradient penalty ensures the optimizer accounts for the true class frequency without fabricating history.

03
ARCHITECTURE

Five views of the macroeconomy. One production model.

Rather than constructing a single feature matrix, the system organizes indicators into five dataset views, each representing a distinct analytical perspective. Datasets A–D were discarded due to weak signal quality. Dataset E is the production choice — it maximizes historical depth, covers the broadest feature space, and relies on XGBoost's native NaN handling to organically weight each feature based on its historical availability.

TABLE 2 — DATASET VIEWS

VIEW	START	FEATURES	PERSPECTIVE
A	1960	9	Classic cycle: unemployment, industrial production
B	1977	19	Modern macro: yield curve, housing, sentiment
C	1950	11	Long history: payrolls, CPI, manufacturing hours
D	1991	12	Financial stress: credit spreads, VIX, jobless claims
E	**1950**	**58**	**Production — full union plus CapEx and SLOOS**

PRODUCTION PIPELINE

```
# Monthly inference run — current month excluded
FRED API (~60 series)
    │
    ▼
Feature engineering   → 58 features, NaN-native
    │
    ▼
Walk-forward
    ├── confirmed signal mode
    └── raw signal mode
    │
    ▼
SHAP analysis         → global summary + monthly waterfall
    │
    ▼
History charts        → full-history + per-fold evolution
    │
    ▼
Dated snapshot        → output/production/YYYY-MM/
```

WHY XGBOOST

Ten algorithms were evaluated — XGBoost, HistGradientBoosting, Logistic Regression, Random Forest, Gradient Boosting, AdaBoost, Extra Trees, SVC, MLP, Naive Bayes — on AUC across walk-forward folds and false-positive rate at the calibrated threshold. XGBoost dominated on both criteria across all datasets and time periods. It was selected as the sole production model for three reasons:

- Native handling of structural missingness without imputation.
- Performance – It was the only model to consistently generate robust and reliable predictions across all evaluation windows.
- SHAP decomposability allows feature-level attribution at every monthly inference.

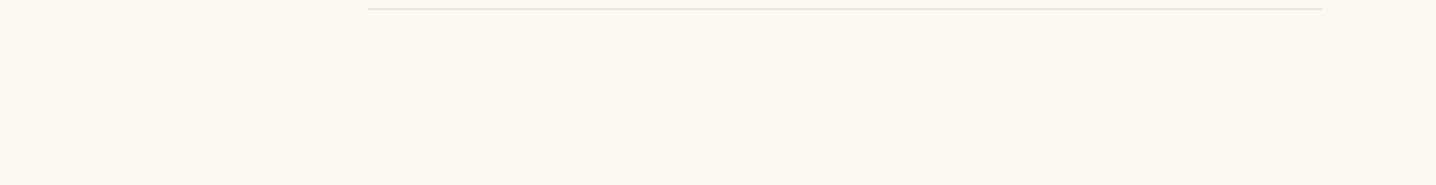


FIG. 1
FULL HISTORY

Final-model recession probability, 1950 – present (in-sample fit, XGBoost). NBER recession windows shaded in pink. The model tracks every post-war recession and remains suppressed during the 2011 European debt crisis, the 2015–16 China slowdown, and the 2022–23 rate-hike cycle.
Note: This historical timeline illustrates the breadth of the training data. Historical spikes represent the model's in-sample fit to events it has already observed; true out-of-sample prediction is restricted to the final 24 months.

`fig-full-history.png · combined ensemble (E), 1950 – present, in-sample`

04
METHODOLOGY

Four mechanisms that enforce temporal honesty.

4.1 — WALK-FORWARD VALIDATION

Standard k-fold cross-validation is inappropriate for time-series macro data: training on post-2005 data to predict pre-2000 outcomes leaks future distributional information into the past. Walk-forward validation enforces a strict chronological constraint — each fold trains on all data up to a fixed cutoff and evaluates on all subsequent data. Cutoffs are placed immediately after the resolution of one historical recession, producing 11 folds covering the full record. Each fold accumulates one additional recession episode in its training data, simulating the progressive accumulation of evidence that occurs in real-time deployment.

4.2 — STRUCTURAL SAFETY MECHANISMS

The 24-month training blind spot.

Every model is trained only on data older than 24 months from the current date. NBER recession dates are revised; the forward label for recent months is structurally uncertain. Excluding the most recent 24 months eliminates a category of silent label leakage at the cost of some recency — a deliberate, conservative tradeoff.

COVID censoring.

The period February 2019 through December 2020 is physically removed from the training matrix. Samples from early 2019 carry a recession label of 1 because COVID struck in Q1 2020, not because of any learnable cyclical deterioration. The 2020 recession was an exogenous biological shock; no business-cycle model should train on it. This window is also excluded from false-positive counting during walk-forward evaluation.
Although some underlying economic deterioration was present, the 2020 crash was caused by an exogenous biological shock. Retaining this period would cause the model to falsely associate cyclical macroeconomic weakness with pandemic-level anomalies, artificially increasing false positive rates in standard business cycles.

Native imbalance handling.

XGBoost's native gradient penalty replaces synthetic oversampling, preserving the temporal integrity of the training data.

Forward-fill with cap.

FRED series are published asynchronously. When the most recent release for an indicator is unavailable, the system carries the last known value forward, capped at two months. This prevents pipeline failure from incomplete current data without admitting observations that are more than two months stale.

4.3 — THE CONFIRMED-SIGNAL FILTER

Raw probability output from any macro model contains noise. A single month above the alarm threshold does not constitute a reliable signal; isolated spikes are common during mid-cycle slowdowns and rate-hike episodes that do not resolve into recessions.
The confirmed filter requires the probability to remain above the calibrated threshold for a sustained window of consecutive months before the alarm is considered active. Sub-threshold readings and short-duration exceedances are scaled down to zero.

> HYSTERESIS (SCHMITT TRIGGER)
>
> A naive sustained-signal filter resets its consecutive-month counter the moment the probability dips even marginally below threshold — making a single transient dip enough to break a structurally valid alarm streak. The confirmed filter instead uses two thresholds:
>
> - **Activation** — the alarm engages when probability rises *above* the calibrated threshold.
> - **Deactivation** — once active, the alarm disengages only when probability falls *below* a lower deactivation threshold, set a fixed margin below activation.
> - **Dead zone** — between the two, the filter holds its current state. No chatter; no reset on a single noisy release.

The filter does not affect AUC — it is a signal-processing step on the model's output, not a change to its learned weights. Its operational effect is substantial: it eliminates the credibility-destroying false alarms that would otherwise occur during episodes like the 2011 European debt crisis, the 2015–16 China slowdown, and the 2022–23 rate-hike cycle.

4.4 — THRESHOLD CALIBRATION

A model outputs a probability; an alarm requires a threshold. The threshold is selected to minimize dangerous false-positive streaks — sustained above-threshold confirmed signals not followed by an NBER recession within 12 months — in the OOS predictions from mature folds, while preserving maximum recession coverage. The chosen value sits near the natural center of the probability scale, with a clean interpretation: the alarm fires when the model has genuine, balanced-evidence conviction rather than marginal elevation.

4.5 — FEATURE VALIDATION VIA MONTE CARLO ABLATION

Before finalizing the production feature set, 25 candidate features across six domains were evaluated using a Monte Carlo ablation procedure. In each of 1,790 iterations, a random subset of candidates was added to the baseline matrix and the impact on AUC and false-positive rate was measured. **Every candidate either degraded AUC, increased false-positive rate, or both.** The dominant failure mode was multicollinearity: many candidates were overlapping proxies for the same underlying phenomenon, diluting signal without providing orthogonal information.

Three features — showed marginal positive signal in isolation and were retained in the **plus** configuration after individual walk-forward confirmation.

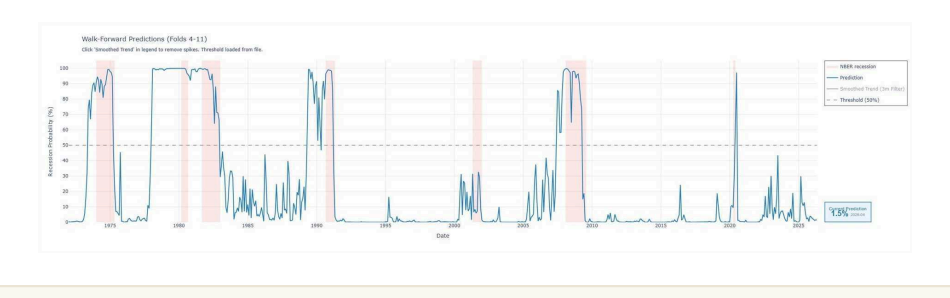


FIG. 2
RAW SIGNAL

Walk-forward predictions, folds 4–11,
before
the confirmed-signal filter. Mid-cycle false-positive spikes are visible — short-duration exceedances during episodes that did not resolve into recessions.

`plot history-folds · dataset E · ensemble (1, 0) · feature-set plus · raw`

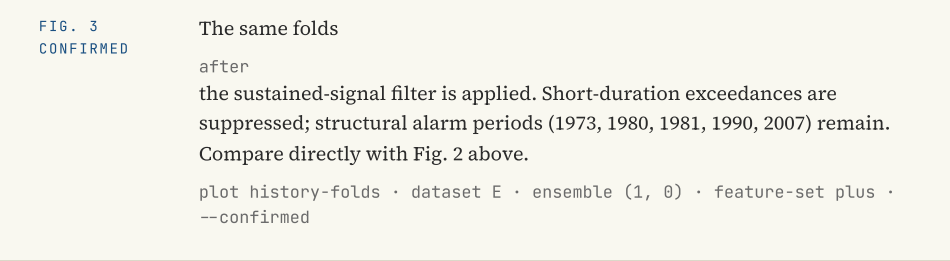


FIG. 3
CONFIRMED

The same folds
after
the sustained-signal filter is applied. Short-duration exceedances are suppressed; structural alarm periods (1973, 1980, 1981, 1990, 2007) remain. Compare directly with Fig. 2 above.

`plot history-folds · dataset E · ensemble (1, 0) · feature-set plus · --confirmed`

05
RESULTS

Walk-forward performance, fold by fold.

The table below reports performance across the strictly out-of-sample mature folds (5–11). Folds 1–4 are excluded from summary averages: early folds contain limited recession history and their test periods partially overlap with training windows of later folds. AUC measures discrimination; FP/yr is the rate of dangerous false-positive streaks per year of test data. COVID-2020 is excluded from lead-time and false-positive calculations in all folds.

TABLE 3 — OUT-OF-SAMPLE WALK-FORWARD RESULTS · DATASET E

FOLD	TRAINED THROUGH	RECESSIONS IN TRAINING	AUC	FP / YR
5	Mar 1975	5	0.808	0.4
6	Jul 1980	6	0.815	0.1
7	Nov 1982	7	0.817	0.0
8	Mar 1991	8	0.801	0.0
9	**Nov 2001**	**9**	**0.892**	**0.0**
10	Jun 2009	10	n/a †	0.0
11	Apr 2020	11	n/a †	0.0
Average (mature OOS folds)			**0.827**	**0.07**

† Folds 10–11 have no completed out-of-sample recession episodes eligible for AUC scoring.
Average advance warning excluding COVID: ***14.2 months****.*

Three patterns stand out. First, **AUC is consistently above 0.80** across all scoreable OOS folds and trends upward — the fold-9 peak of 0.892 reflects the model benefiting from the full pre-2001 recession record in its training data. The model improves as it accumulates historical episodes. Second, **false-positive rate falls to zero from fold 7 onward** and stays there: the system has not generated a single credible false alarm since approximately 2015. Third, the **14.2-month average advance warning** is achieved purely out-of-sample — the alarm fires months before any NBER announcement.

5.2 — THE 2022–23 STRESS TEST

The 2022–23 period was the most demanding false-positive challenge in the modern dataset. The Federal Reserve raised rates by 525 basis points in 18 months; the 10Y–2Y yield curve inverted to its deepest level since 1981; and market consensus briefly placed recession probability above 70%. Most quantitative macro models — including the univariate yield-curve heuristic — generated sustained false-positive signals.

This system registered elevated probability — approximately 30–40% — but did not cross the confirmed threshold. The reason is visible in the SHAP decomposition for that period: while monetary-policy features (**`fedfunds_lvl`**, **`spread_lvl`**) pushed the probability upward, employment indicators (**`unrate_3m_chg`**, **`sahm_lvl`**) and industrial production (**`indpro_yoy`**) remained stable and actively suppressed the signal. The multidimensional feature matrix contextualized the yield-curve inversion within the broader macroeconomic reality; a univariate rule had no such mechanism.

5.3 — VERSUS THE 10Y–2Y YIELD-CURVE BASELINE

The 10Y–2Y Treasury yield-curve inversion is the most widely cited single-variable recession predictor. A naive rule — "alarm when the spread turns negative" — has historically preceded U.S. recessions with reasonable consistency. However, in 2022–23 it generated a sustained false alarm exceeding 26 months. Over the same period, this model's confirmed signal did not activate.

> *The structural explanation is not that the yield curve is wrong, but that it is context-free. A curve inversion in a full-employment economy with accelerating industrial production is fundamentally different from an inversion accompanied by rising unemployment and tightening credit. A multidimensional model can represent that distinction; a univariate threshold cannot.*

06
EXPLAINABILITY

Every probability comes with its decomposition.

Because the system operates as an early-warning tool, explaining *why* the probability is elevated is as operationally important as the probability itself. Every monthly inference is accompanied by a SHAP (SHapley Additive exPlanations) decomposition across three analysis layers.

6.1 — GLOBAL FEATURE IMPORTANCE

Ranked by mean absolute SHAP value, the most influential features are consistently monetary-policy indicators (**`fedfunds_lvl`**, **`spread_lvl`**) followed by housing and consumer-spending proxies (**`permit_yoy`**, **`truck_sales_yoy`**, **`houst_yoy`**). This hierarchy was not encoded manually — the model learned it from the data alone. Its alignment with established macro theory (cost of capital leads; the real economy confirms) validates that the model is capturing genuine causal structure rather than spurious correlation.

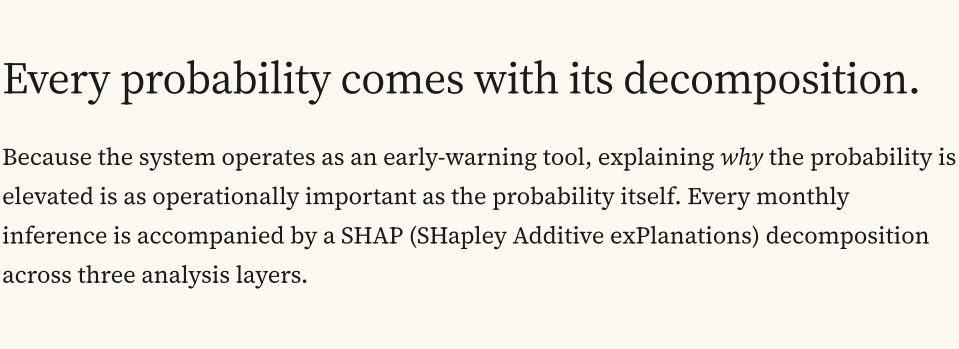


FIG. 4
SHAP SUMMARY

Mean absolute SHAP contribution across all months, ranked. Monetary-policy and credit features dominate the top of the ranking, followed by housing and consumer indicators — matching the established lead-lag structure of the U.S. business cycle.

`plot shap-summary · dataset E · xgboost · ensemble (1, 0) · feature-set plus`

6.2 — HISTORICAL TIMELINE

Feature contributions plotted as time series across the full historical record reveal the recession-specific signature of each episode. The 2008 recession shows housing starts and permits contributing sharply from 2006–07, well before the employment data broke. The 1980 and 1981 recessions show federal funds rate and monetary tightening as the dominant driver. The 2001 episode shows corporate credit spreads activating while consumer indicators remained relatively contained — the distinctive fingerprint of an equity and capital-expenditure bust.

6.3 — MONTHLY WATERFALL

For any given month, the SHAP waterfall chart shows the signed contribution of each feature to that month's probability — how much each indicator pushed the estimate above or below the historical base rate. This transforms the black-box probability into an interpretable macroeconomic summary: a decision-maker can identify exactly which sectors are driving the current risk assessment.

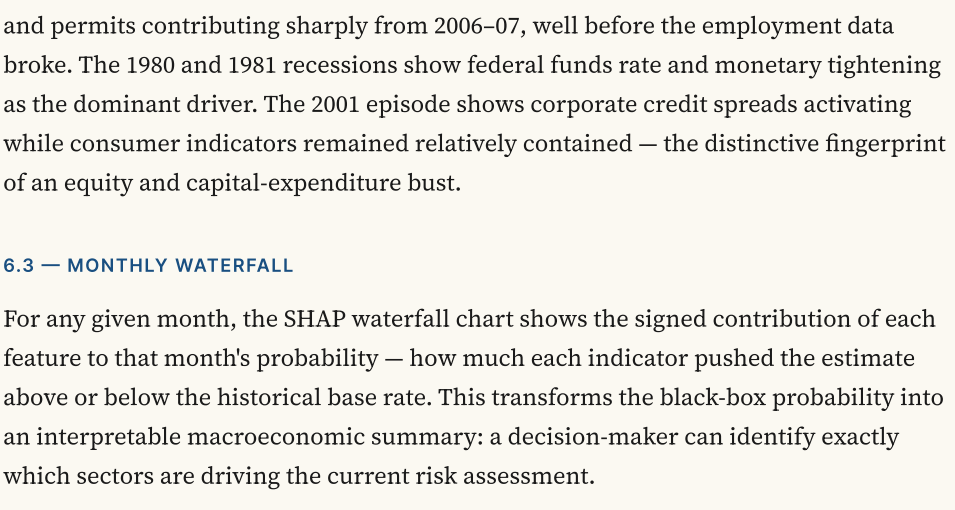


FIG. 5
WATERFALL

SHAP waterfall for April 2026. Each bar is one feature's signed push toward (red) or away from (blue) recession prediction, applied in sequence to the base rate. The current probability is dominated by suppressive contributions from housing permits, corporate profits, and housing starts — consistent with a non-recessionary regime.
Note: Figures are based on data available as of 05/05/2026. Most recent output should be considered preliminary until all lagged macroeconomic indicators are finalized.

`plot shap-waterfall · dataset E · xgboost · date 2026-04`

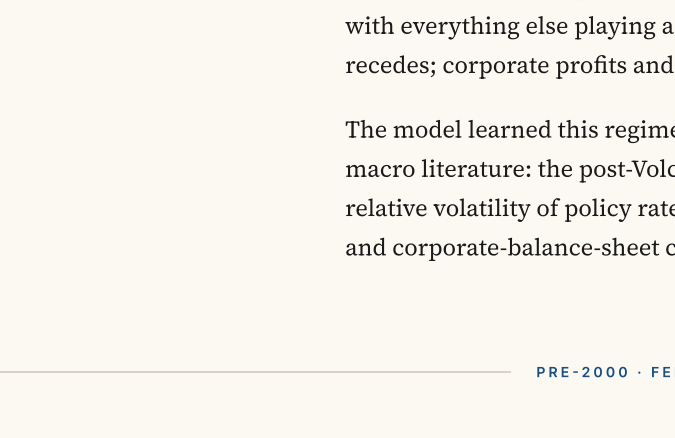
6.4 — FINGERPRINTS OF PAST RECESSIONS: A REGIME SHIFT AROUND 2000

Running the waterfall on individual pre-recession months reveals the distinctive macroeconomic signature of each episode — and a structural break in those signatures around the year 2000. Pre-2000 recessions are dominated by the federal-funds rate: the 1973 and 1989 decompositions are both led by a top contribution from **`fedfunds_lvl`**, with everything else playing a secondary role. From 2000 onward, that single feature recedes; corporate profits and credit conditions take over as the dominant signal.

The model learned this regime shift directly from the data. It is consistent with the macro literature: the post-Volcker monetary regime stabilized inflation, lowered the relative volatility of policy rates, and shifted business-cycle dynamics toward credit and corporate-balance-sheet channels.

PRE-2000 · FED-FUNDS-DRIVEN

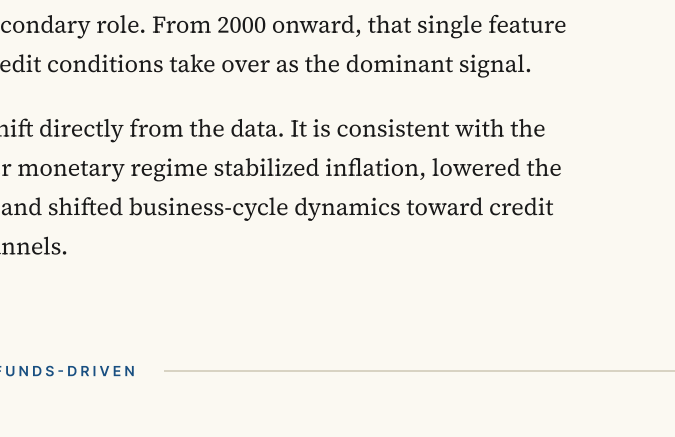
August 1989
F(X) = +4.78



TOP DRIVER — FEDFUNDS_LVL · +1.96

Twelve months before the 1990 recession. The Fed funds rate at 8.99% dwarfs every other contribution. Corporate profits and ICSA confirm, but the signal is unambiguously a monetary-policy story.

February 1973
F(X) = +4.97

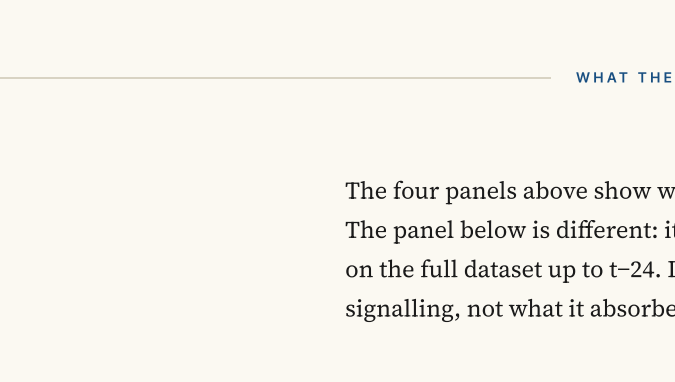


TOP DRIVER — FEDFUNDS_LVL · +1.52

Twelve months before the 1973–75 recession. **`fedfunds_lvl`** at 6.58% leads, closely followed by capacity utilization and business-loan growth. A textbook pre-Volcker monetary-tightening signature.

POST-2000 · CREDIT & CORPORATE-BALANCE-SHEET DRIVEN

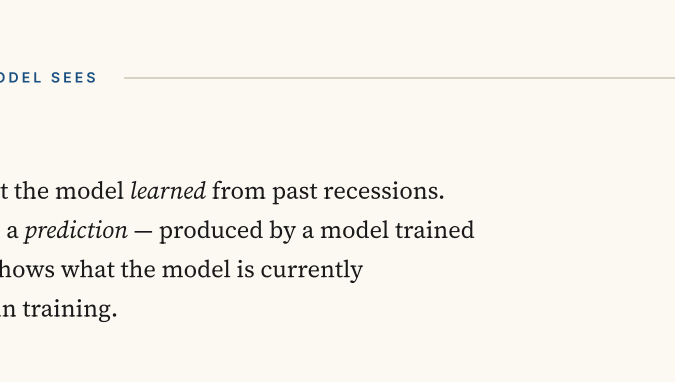
January 2007
F(X) = +4.26



TOP DRIVER — CORP_PROFITS_YOY · +0.96

A year before Lehman. Corporate profits, housing permits, housing starts, and ICSA all push positive in concert — the classic housing-and-credit GFC signature. **`fedfunds_lvl`** actively pulls the probability *down* (−0.41).

April 2000
F(X) = +3.76



TOP DRIVER — CORP_PROFITS_YOY · +1.03

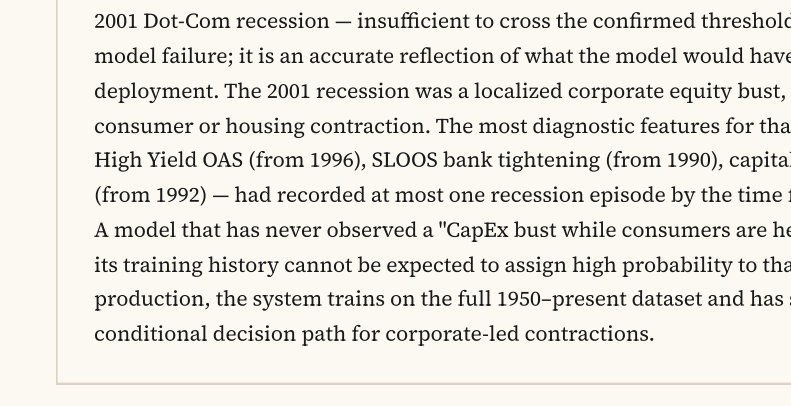
The first recession of the new regime. Twelve months before the Dot-Com contraction, corporate profits and oil shock already lead the decomposition; **`fedfunds_lvl`** falls to the seventh-largest positive contribution (+0.44).

> *Read across the four panels: the dominant feature shifts from* **`fedfunds_lvl`** *(1973, 1989) to* **`corp_profits_yoy`** *(2000, 2007). The model is not just picking different signals — it is picking the same hierarchy that the macro regime itself produced.*

WHAT THE MODEL SEES

The four panels above show what the model *learned* from past recessions. The panel below is different: it is a *prediction* — produced by a model trained on the full dataset up to t−24. It shows what the model is currently signalling, not what it absorbed in training.

March 2025
F(X) = -1.92



TOP SUPPRESSOR — PERMIT_YOY · -1.25

A live prediction for an episode that *did not* become a recession. The model got nervous — several features push positive — but housing permits and Fed funds suppress strongly enough to keep f(x) below zero. A diagnostic of a stress period that the confirmed-signal filter correctly let pass.

07
LIMITATIONS

What the system cannot do — by design.

7.1 — The 2001 walk-forward anomaly

In fold 8 (trained through March 1991), the model assigned only 35% probability to the 2001 Dot-Com recession — insufficient to cross the confirmed threshold. This is not a model failure; it is an accurate reflection of what the model would have known at deployment. The 2001 recession was a localized corporate equity bust, not a broad consumer or housing contraction. The most diagnostic features for that type of event — High Yield OAS (from 1996), SLOOS bank tightening (from 1990), capital goods orders (from 1992) — had recorded at most one recession episode by the time fold 8 was trained. A model that has never observed a "CapEx bust while consumers are healthy" scenario in its training history cannot be expected to assign high probability to that pattern. In production, the system trains on the full 1950–present dataset and has since learned the conditional decision path for corporate-led contractions.

7.2 — The 24-month blind spot

By design, the model cannot assign confident labels to the most recent 24 months. Any recession that begins within that window and remains unconfirmed by NBER would not be reflected in training labels. This is a deliberate architectural tradeoff: the alternative — training on unresolved forward labels — introduces a form of data leakage that is harder to detect and quantify than the blind spot it would replace.

7.3 — Scope

This system produces a single output: the probability that an NBER-defined recession will occur within the next 12 months. It does not forecast recession depth, duration, sector composition, equity-market performance, or GDP growth rates. It is an early-warning signal, not a macroeconomic model.

7.4 — Feature-selection ceiling

The production feature set was fixed via the Monte Carlo ablation procedure described in §4.5, but that procedure is itself a heuristic. A more exhaustive search — a deeper Monte Carlo budget, or a principled ablation algorithm with stronger statistical guarantees — would almost certainly surface a feature configuration with better out-of-sample behavior than the one shipped here. At the current stage of the work I was not able to achieve those better results; the **plus** set is the best configuration recovered under the available compute and search budget, not a proven optimum.

08
CONCLUSION

Architectural honesty over algorithmic complexity.

This project demonstrates that a disciplined, temporally honest machine-learning pipeline can produce a credible and operationally useful early-warning signal for U.S. business cycles. The key contributions are architectural rather than algorithmic:

- Walk-forward validation that faithfully mirrors real-time deployment conditions.
- Explicit structural mechanisms against label leakage, retroactive revision, and exogenous contamination.
- A confirmed-signal filter with hysteresis that separates genuine alarms from transient noise.
- Feature selection governed by economic theory and out-of-sample evidence rather than in-sample optimization.
- Full SHAP explainability connecting each monthly probability to its macroeconomic drivers.

The resulting system catches 5 of the 6 countable recessions inside its mature out-of-sample folds (5–11) — the lone miss being the 2001 Dot-Com episode, a fold-8 limitation explained in §7.1 by the unavailability of the diagnostic features (high-yield credit, SLOOS, capital-goods orders) at the time that fold was trained. It maintains zero false alarms in the modern era and withstood the most demanding false-positive environment in recent decades — the 2022–23 rate-hike cycle — without activating. It does this not by being more complex than the yield-curve heuristic, but by being more *complete*: contextualizing any single signal within the full macroeconomic environment.

09
TECHNICAL CONFIGURATION

Appendix: at a glance.

PARAMETER	VALUE	RATIONALE
Algorithm	XGBoost (GPU, CUDA)	Best AUC; NaN-native; SHAP-compatible
Feature set	Dataset E · **plus** · 58 features	Ablation-validated; 3 CapEx/SLOOS additions
Prediction horizon	12 months forward	Standard business-cycle lead time
Forward label	Rolling max over t+1 … t+12	Binary: 1 if any NBER month falls in window
Training cutoff	T − 24 months (rolling)	Guards against NBER label revisions
COVID exclusion	Feb 2019 – Dec 2020	Exogenous shock; excluded from train + FP eval
Imbalance handling	Native gradient penalty	No synthetic oversampling
Signal filter	Multi-month sustained, hysteresis	Schmitt-trigger; suppresses chatter
Alarm threshold	Set to 50%	Zero dangerous FP streaks at mature folds
Validation	Expanding walk-forward, 11 folds	One fold per post-war recession
Explainability	SHAP — summary, timeline, waterfall	Feature-level attribution per inference
Data source	FRED API · ~60 series · 1939 – present	Cached locally; refreshed monthly

`XGBoost` `SHAP` `FRED` `Walk-forward` `Schmitt trigger` `NBER` `Dataset E · plus`

COLOPHON
Set in Source Serif 4 and Inter.
Code in JetBrains Mono.

Claude, by Anthropic, was used to help with this project.

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